HW 3

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Link to git repository for code.

Problem 1

Part 1

Since A is a square matrix, its characteristic polynomial is defined as follows.

$$p_A(\lambda) = \det(A - I\lambda)$$

The term $A - I\lambda$ is supposed to result in a singular matrix, so the determinant in the equation above is supposed to equal zero. The diagonals of $A - I\lambda$ are in the form $a_{ii} - \lambda$.

Because A is a square matrix, its eigenvalues are the diagonals of its upper triangular matrix factor, and the determinant of an upper triangular matrix, obtaind through gaussian elimination (GE), the product of its diagonal. The equation is as follows.

$$\det(GE(A - I\lambda)) = (u_{i,i} - \lambda)(u_{i+1,i+1} - \lambda)...(u_{n,n} - \lambda)$$

To ensure that the determinant is non-zero, set $\lambda = 0$. Additionally, since the diagonals of the upper triangular matrix are the eigen values of A, $u_{i,i} = \lambda_i$. The resulting equation is:

$$p_A(0) = \det(GE(A - I(0))) = (\lambda_i)(\lambda_{i+1})...(\lambda_n) = \prod_{i=1}^{n} \lambda_i$$

Part 2

The iteration matrix is defined as $B_w = (D + wL)^{-1}[(1 - w)D - wU]$. By taking the determinant of B_w, the following expression is formed.

$$\det(B_w) = \det(D + wL)^{-1}[\det((w - 1)D - wU)] = \det(D)^{-1}[\det((1 - w)D)]$$

Using part 1, the determinant of D is the product of its diagonal entries or eigenvalues.

$$\det(B_w) = \det(D)^{-1}[\det((1 - w)D)] = \prod_{i=1}^{n} \frac{1}{\lambda_i} \prod_{i=1}^{n} (1 - w)\lambda_i = \prod_{i=1}^{n} \frac{(w - 1)\lambda_i}{\lambda_i} = \prod_{i=1}^{n} (1 - w)$$

Since the determinant is the product of a square matrices eigenvalues, the eigenvalues of are $\Lambda(B_w) = (1 - w)$. The spectral radius, $\rho(B_w)$ is bounded below by the gemoetric mean of the eigen values, $\sqrt[n]{\prod_{i=1}^{n} |\lambda_i|}$, so $\rho(B_w) \geq \sqrt[n]{|(w - 1)|^n}$; or more concisely as $\rho(\mathbf{B_w}) \geq |\mathbf{w - 1}|$.

Problem 2

Part 1

$$B_J = \begin{pmatrix} 0 & -1/3 & 0 \\ -1/3 & 0 & -1/3 \\ 0 & -1/3 & 0 \end{pmatrix}$$

$$B_{GS} = \begin{pmatrix} 0 & -1/3 & 0 \\ 0 & 1/9 & -1/3 \\ 0 & -1/27 & 1/9 \end{pmatrix}$$

The infinity norm of B_J is $\|B_J\|_\infty = \max(|-1/3|, |-1/3| + |-1/3|, |-1/3|) = 2/3$. $\rho(B_J) = \max_\lambda(\lambda(\lambda^2 - 2/9) = 0) = \frac{\sqrt{2}}{3}$. The infinity norm of B_{GS} is $\|B_{GS}\|_\infty = \max(|-1/3|, |1/9| + |-1/3|, |-1/27| + |1/9|) = 4/9$. $\rho(B_{GS}) = \max_\lambda(\lambda(\lambda^2 + 2/9\lambda + 2/81) = 0) = 2/9$.

The optimal ω for SOR is $\frac{2}{1+\sqrt{1-\rho(B_J)^2}} = 9 - 3\sqrt{7}$. This only works because A is tridiagonal and strictly diagonally dominant, ie.) symmetric positive definite.

Part 2

Tables

Table 1: Jacobi Method Iterations

k	x_1	x_2	x_3	e	e_{suc}
1	1.333333	1.666667	1.333333	0.666667	0.000000
2	0.777778	0.777778	0.777778	0.222222	0.333333
3	1.074074	1.148148	1.074074	0.148148	0.666667
4	0.950617	0.950617	0.950617	0.049383	0.333333
5	1.016461	1.032922	1.016461	0.032922	0.666667
6	0.989026	0.989026	0.989026	0.010974	0.333333
7	1.003658	1.007316	1.003658	0.007316	0.666667
8	0.997561	0.997561	0.997561	0.002439	0.333333
9	1.000813	1.001626	1.000813	0.001626	0.666667
10	0.999458	0.999458	0.999458	0.000542	0.333333

Table 2: Gauss–Seidel Method Iterations

k	x_1	x_2	x_3	e	e_{suc}
1	1.333333	1.222222	0.925926	0.333333	0.000000
2	0.925926	1.049383	0.983539	0.074074	0.222222
3	0.983539	1.010974	0.996342	0.016461	0.222222
4	0.996342	1.002439	0.999187	0.003658	0.222222
5	0.999187	1.000542	0.999819	0.000813	0.222222
6	0.999819	1.000120	0.999960	0.000181	0.222222
7	0.999960	1.000027	0.999991	0.000040	0.222222
8	0.999991	1.000006	0.999998	0.000009	0.222222
9	0.999998	1.000001	1.000000	0.000002	0.222222
10	1.000000	1.000000	1.000000	0.000000	0.222222

Table 3: SOR Method Iterations ($\omega = 9 - 3\sqrt{7} \approx 1.1707$)

k	x_1	x_2	x_3	e	e_{suc}
1	1.416995	1.269275	0.967356	0.416995	0.000000
2	0.878445	1.037729	0.988683	0.121555	0.291503
3	0.994262	1.003675	0.999408	0.005738	0.047208
4	0.999058	1.000313	0.999926	0.000942	0.164097
5	0.999948	1.000025	0.999996	0.000052	0.054849
6	0.999994	1.000002	1.000000	0.000006	0.107079
7	1.000000	1.000000	1.000000	0.000000	0.057452
8	1.000000	1.000000	1.000000	0.000000	0.091118
9	1.000000	1.000000	1.000000	0.000000	0.058765
10	1.000000	1.000000	1.000000	0.000000	0.083607

Discussion

The jacobi method, while it minimizes the error, does not get the lowest possible error. Gauss-Seidel reaches the lowest error at the last iteration. For the SOR method, the solution converge smuch earlier than the Gauss-Seidel method.

Problem 3

Part 1

$$x^{(k+1)} = Q^{-1}(Q - A)x^{(k)} + Q^{-1}b \rightarrow x^{(k+1)} = x^{(k)} - AQ^{-1}x^{(k)} + Q^{-1}b$$

$$z^{(k)} = Q^{-1}(b - Ax^{(k)})$$

$$x^{(k+1)} = x^{(k)} + Q^{-1}(b - Ax^{(k)}) \rightarrow x^{(k+1)} = x^{(k)} + z^{(k)}$$

From the equations above, the iteration process is equuivalent.

Part 2

proof for z

$$z^{(k+1)} = Q^{-1}r^{(k+1)} = Q^{-1}(b - Ax^{(k+1)}) = Q^{-1}(b - A(x^{(k)} + z^{(k)})) = Q^{-1}(b - Ax^{(k)} - Az^{(k)}) =$$
$$Q^{-1}(r^{(k)} - Az^{(k)}) = Q^{-1}(Qz^{(k)} - Az^{(k)}) = (I - Q^{-1}A)z^{(k)}$$

proof for r

$$r^{(k+1)} = b - Ax^{(k+1)} = b - A(x^{(k)} + z^{(k)}) = b - Ax^{(k)} - Az^{(k)} = r^{(k)} - AQ^{-1}r^{(k)} = (I - AQ^{-1})r^{(k)}$$

Problem 4

Table 4: Comparison of Jacobi and Gauss–Seidel Convergence

Method	k	r_{norm}	$r_{\text{successive}}$
Jacobi	9	0.007820	0.500488
Gauss–Seidel	5	0.009747	0.502953

Discussion

The results of the table show that the Gauss-Siedel method converges much faster than the Jacobi. This is due to the fact that the spectral radius of the Gauss-Seidel matrix is the spectral radius of the jacobi matrix squared. Since both spectral radii are less than one, the Gauss-Seidel method converges much faster. Therefore, the behavior in the table above matches what we've learned in class.